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Survey Paper on

the security and privacy risks of the Internet of Medical Things (IoMT)

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A Comprehensive Survey on Security and Privacy Challenges in Internet of Medical Things Applications: Deep Learning and Machine Learning Solutions, Obstacles and Future Directions

Survey Paper Link: https://ieeexplore.ieee.org/stamp/stamp.jsp?arnumber=11079573

Summary

The survey reviews security and privacy problems in the Internet of Medical Things (IoMT). In IoMT, connected wearables, implantable sensors, hospital equipment, networks and cloud services support remote monitoring and data driven care. These tools can improve diagnosis, treatment and access to care. But their many designs and limited computing power make them hard to secure. Cyberattacks can expose sensitive health data, disrupt services, or put patients at risk.

The authors group threats into three IoMT layers. At the perception layer, risks include device tampering, sensor spoofing and side channel attacks. At the network layer, threats include eavesdropping, man in the middle attacks, replay attacks and denial of service. At the application layer, risks include ransomware, phishing, session hijacking and database attacks.

The survey covers defenses such as lightweight encryption, secure authentication, access controls, physical unclonable functions, blockchain and secure software updates. It also discusses machine learning and deep learning for anomaly detection and threat identification. Federated learning can train shared models while reducing patient data sharing.

The paper concludes that no single method can secure IoMT. Protection needs layered, adaptable defenses that balance security with energy use, realtime response, scalability, interoperability and regulatory compliance. Future work should strengthen privacy preserving data sharing, AI resilience, device standards and continuous monitoring. Trustworthy IoMT requires coordinated effort from researchers, healthcare providers, manufacturers and policymakers.

Major Contributions of the Survey

The survey's main value is that it gives a complete view of IoMT security. Instead of looking at one technology or one kind of attack on its own, it pulls together many connected parts. It covers the IoMT architecture, security needs, types of attacks, machine learning and deep learning methods and ways to defend against threats. The goal of the analysis is to help

researchers and practitioners see how these pieces fit together. It also highlights where current protections are still not strong enough.

1. A layered view of IoMT architecture and threats

The paper explains the Internet of Medical Things (IoMT) using three layers:

Perception layer: Medical sensors and devices collect information about patients and their surroundings. For example, a heart monitor may track a patient's pulse.

Network layer: Gateways and communication systems send information between devices and computers that process it.

Application layer: Cloud services and healthcare apps store, analyze and display the information.

The paper groups security threats by the layer where they can happen. At the perception layer, attackers might tamper with a device, use side channel attacks, fake a sensor's location, or add a harmful device to the system. At the network layer, threats include secretly listening to data, intercepting and changing messages, replaying old messages, or disrupting services. At the application layer, threats include ransomware, phishing, taking over a user's session, cross site scripting, repeated password guesses and SQL injection.

This way of grouping threats connects each risk to the parts of the system and the information it affects. It also shows that security must protect more than the medical devices themselves. Threats can arise when information is collected, sent, stored, or processed by an app.

2. A consolidated account of IoMT security and privacy requirements

The survey explains the main goals of security for the Internet of Medical Things (IoMT). These goals include keeping information private, preventing data from being changed, resisting attacks, controlling who can access systems, tracking activity and protecting personal information.

These goals reflect the real needs of healthcare. Confidentiality keeps sensitive records away from people who are not allowed to see them. Data integrity helps make sure patient measurements are accurate and unchanged. Systems must also stay available and recover from problems, because an interruption could delay urgent care.

The survey also describes why securing IoMT devices is difficult. Many devices have limited computing power, memory and battery life. At the same time, healthcare systems need quick responses and must work with equipment from different companies and institutions.

Other challenges include supporting many devices, updating them, meeting regulations, controlling costs and energy use and limiting environmental harm. The survey presents security as something that must be considered across the whole system—not just by choosing a strong encryption method.

3. A broad review of defensive technologies

The survey looks at several ways to protect Internet of Medical Things (IoMT) systems, such as connected medical devices:

Simple, efficient encryption and secure connections can protect data while it is being sent or stored. These methods must also work within the limits of small or lowpowered devices.

Authentication and access controls help make sure only the right people and devices can access a system. Examples include logging in with more than one form of proof, giving users access based on their roles and using unique physical features to identify a device.

Blockchain tools can keep records of transactions, help show that data has not been changed and manage who is allowed to access information.

Artificial intelligence and machine learning can monitor devices and networks. For example, they may flag unusual behavior or unexpected network traffic.

Federated learning lets organizations train a shared AI model without gathering all patient data in one central place. Each organization can keep its data locally and share lessons learned by the model.

Secure updates and ongoing monitoring help protect devices as new weaknesses and threats emerge.

The review emphasizes that these methods do different jobs and have different strengths and limits. Encryption, for example, can keep information private and help prevent changes. Machinelearning tools may spot unusual activity. Blockchain can provide a record that people can check, but it may not detect an attack as it happens.

For this reason, the survey recommends combining methods that complement one another instead of relying on just one.

4. Examination of machine learning and deep learning methods

The paper focuses on how machine learning (ML) and deep learning (DL) can help protect the Internet of Medical Things (IoMT). These methods can analyze large amounts of medical

and network data. They can spot unusual activity, help verify users and devices and identify possible attacks.

The paper reviews several methods, including neural networks, autoencoders and federated learning. For example, these systems might notice when a medical device sends data in an unusual pattern.

The paper also explains the limits of AIbased security. Some models need a lot of computing power, are hard to understand , or do not work well on low power devices. Their results can also be affected by attackers who manipulate the system or by corrupted data used during training.

These risks matter especially in healthcare. A wrong result from an AI system could affect patient care, not just cause a technical problem. The paper highlights explainable AI, stronger training methods and trustworthy AI as important topics for future research.

5. Identification of research gaps and future directions

The survey finds several gaps in research on the security of Internet of Medical Things (IoMT) devices. Many solutions do not work well with different types of devices, detect threats quickly, or use little enough energy for devices with limited power.

Security measures often protect privacy, verify users, or keep data accurate separately. They do not always bring these protections together in one system. Other challenges include making devices work together, adding security to large systems, updating devices safely, protecting privacy when analyzing data and defending AI systems from attacks. For example, attackers might alter the data used to train a medical AI system.

The authors suggest combining computing at the edge—near the devices—with cloud computing. They also recommend creating security tools that use less energy and adapt to new threats, improving standards and device certification and making AI easier to understand and more resistant to attacks. Other priorities include developing encryption that can withstand future quantum computers and designing devices that use energy more efficiently and sustainably.

Finally, they stress that technical security measures should follow healthcare laws and ethical principles.

Overall contribution

The survey brings together research on how the Internet of Medical Things (IoMT) is designed, the threats it faces and the ways to protect it. It also highlights problems that still need to be solved.

The survey explains how threats can affect three parts of an IoMT system: the devices that collect data, the networks that carry it and the applications that use it. For example, an attacker might tamper with a sensor, intercept data sent over a network, or break into a health app. Different security measures can be combined to address these risks.

The main conclusion is that IoMT security must work in layers and adapt to changing risks. It must also account for devices' limited power and computing capacity. Protection should keep patient information safe without slowing devices, delaying urgent updates, disrupting connections between systems, or putting patient safety at risk.

Key challenges

1. Protecting patient privacy and data confidentiality

Connected medical devices manage highly sensitive data, which can include vital signs, diagnoses, medication histories and geographical information. This data may travel through various devices, networks and cloud computing platforms.

If this information lacks adequate protection, it may be accessible to unauthorized individuals. For instance, insufficient encryption, inadequate access controls, or insecure storage can lead to exposed medical records. Privacy can also be compromised if data is collected or shared unnecessarily for care purposes.

To ensure protection for individuals, safeguards must encompass the entire pathway of the information. Access should be restricted to authorized personnel only. When data is processed or distributed, methods that protect privacy should be employed. For instance, identifiable information can be stripped away before a dataset is given to researchers.

2. Securing resource constrained and diverse devices

IoMT devices vary significantly in their processing capabilities, memory size, battery life, operating systems and communication protocols. Many small sensors lack the ability to implement security measures that are typical for traditional computers. Additionally, applying a uniform security strategy across wearables, implants, medical imaging technologies and hospital infrastructures presents challenges. The security measures must be robust enough to withstand attacks while lightweight enough to maintain device functionality and battery performance. Moreover, there must be reliable methods for secure setup, authentication, firmware integrity and software updates for these devices.

3. Detecting and responding to attacks in real time

Threats can target every layer of an IoMT system. At the device layer, attackers may tamper with sensors or spoof readings. During communication, they may eavesdrop, intercept or alter messages, or launch denial of service attacks. At the application layer, ransomware, phishing and software vulnerabilities can disrupt services or expose records. In healthcare, delayed detection can affect clinical decisions or interrupt care. Security systems must therefore identify threats promptly and support practical responses—for example, isolating a compromised device without interrupting essential monitoring.

4. Balancing intelligent detection with reliability and privacy

Machine learning and deep learning techniques can recognize atypical traffic, device activity, or patient data trends. However, they often require significant computational resources and may generate false positives or fail to detect previously unseen threats. Their effectiveness can also decline when the input data varies from what they were originally trained on. The models themselves can be vulnerable to attacks such as data poisoning or adversarial manipulation. In healthcare environments, the rationale behind an alert must be clear and subject to verification. The goal is not only to enhance detection accuracy but also to ensure that a system is reliable, understand able and suitable for the particular clinical situation.

5. Achieving interoperability and scalability

Hospitals and clinics often use devices and software made by many different companies. These devices don't always speak the same language or follow the same safety rules. That makes it hard to share information safely between them.

Also, the more devices you connect, the more chances there are for something to go wrong. Think of it like a house. The more doors and windows you add, the more ways a burglar could try to get in.

A security setup that works well in a small test, like one clinic, might not work once you try to use it across a whole hospital or a large network of care centers. It's a bit like a plan that works fine for a small shop but falls apart when you try to use it for a huge chain of stores.

To fix this, we need a few things. First, everyone should use the same shared rules, so devices can talk to each other safely. Second, networks should be carefully divided into separate sections, so if one part gets attacked, it doesn't spread to everything else. Finally, there needs to be a system that can check who is allowed to access what and that system must be able to grow as more devices and people join.

6. Maintaining secure devices over their full lifecycle

Some medical devices stay in use for many years. But over time, new threats appear and security needs change. Because of this, manufacturers and healthcare providers must plan ahead. They need a plan for finding security weaknesses, fixing them, sending out safe updates over the internet and deciding when a device is too old to keep supporting.

Updating a device is not as simple as updating an app on your phone. Think of it like repairing a plane while it is still flying. Updates must be checked to make sure they are genuine, tested carefully and delivered in a way that does not interrupt patient care.

This means security cannot be an afterthought. It has to be considered at every stage: when the device is first designed, when it is bought, while it is being used, during repairs and updates and even when it is finally retired.

7. Meeting regulatory, operational and cost requirements

Hospitals and clinics need to keep patient information safe. At the same time, doctors and nurses who are allowed to see that information must still be able to get to it quickly. On top of that, these organizations have to follow certain rules about privacy, like HIPAA in the US and GDPR in Europe. Think of these rules as a strict instruction manual for h and ling people's medical details.

But keeping data safe isn't just about rules. It also costs money and staff need time and training to manage it properly. Hospitals also can't let security measures slow down or stop important medical services.

If a security tool is too expensive or too hard to use, staff may stop using it properly, or skip it altogether. So any solution needs to fit smoothly into how doctors and nurses already work. It also needs a way to track who looked at what information and when, so there's a clear record if anyone needs to check later.

List of similar papers and their links

1. Deep Learning Assisted Security and Privacy Provisioning in the Internet of Medical Things Systems: A Survey on Recent Advances (2024)

https://ieeexplore.ieee.org/document/10472482/

2. Security and Privacy for the Internet of Medical Things Enabled Healthcare Systems: A Survey (2019)

https://ieeexplore.ieee.org/document/8936335

3. Survey on Internet of Medical Things Security: A Comprehensive Review (2025)

https://ieeexplore.ieee.org/document/11012798

4. HIDSIoMT: A Deep Learning Based Intelligent Intrusion Detection System for the Internet of Medical Things (2025)

https://ieeexplore.ieee.org/document/10891538

5. Machine Learning and Deep Learning Methods for Intrusion Detection Systems in IoMT: A Survey

https://ieeexplore.ieee.org/document/9738218

6. A Comprehensive and Systematic Literature Review on Intrusion Detection Systems in the Internet of Medical Things: Current Status, Challenges and Opportunities (2025)

Springer / DOI

https://link.springer.com/article/10.1007/s1046202411101w

7. A Survey of Deep Learning Technologies for Intrusion Detection in Internet of Things

https://ieeexplore.ieee.org/document/10379640

Critical Analysis of the IoMT Security and Privacy Challenges

The 2025 study by Nithyavani and Naga Raja looks at security problems across the whole Internet of Medical Things, often shortened to IoMT. That means devices, networks, cloud systems and health apps all working together. The study's biggest strength is how much ground it covers. It brings together types of threats, computer learning methods, data protection tools, blockchain, login security and privacy methods, all in one place. This wide view makes sense because a medical device system can be attacked in many different spots. If you only fix one weak point, the whole system can still be at risk.

But this wide coverage also makes the study's conclusions feel a bit vague. It talks about many tools but doesn't clearly say which ones work best in real hospitals or clinics. For example, it lists smart detection software, blockchain, lightweight security codes and shared learning systems as promising ideas. But these tools solve different problems and come with different costs. Think of it like home security: a motion sensor can tell you someone's in your yard, but it can't lock your door, check who they are, or make sure your valuables are safe. In the same way, a detection tool might spot unusual computer traffic, but it won't confirm a device is genuine, protect the data being sent, or guarantee that medical readings are correct. Blockchain can help you track who changed what, like a shared diary everyone can check. But it doesn't automatically make sure the information written in that diary is true or keep it private. The paper would be stronger if it explained more clearly what each tool actually does, what it assumes and where it falls short.

The study rightly points out that medical devices come in many shapes and sizes and many have very limited power and computing ability. This matters a lot. Security methods built for powerful computers often won't work on tiny, battery run sensors or implants, like a pacemaker. The study also understand s that things like speed, battery life, getting different devices to work together and software updates are all connected problems, not just separate items on a checklist. It doesn't treat security as just add encryption and you're done.

Still, the study sometimes lists these needs side by side instead of showing how they pull against each other. For example, stronger logins or encryption can slow devices down, drain batteries faster systems harder to run. Frequent software updates can improve safety, but if an update fails or causes downtime, it could put a patient's health at risk. A truly useful framework should explain how to balance these trade offs based on what the device does and how much risk a patient faces. A fitness tracker, a heart monitor you wear at home and an implanted device should not be judged by the exact same rules, because the stakes are very different.

The study also sorts threats into layers, like floors in a building, which is a helpful way to organize ideas. But in real attacks, trouble doesn't always stay on one floor. For example, if a hacker breaks into an employee's account, they might reach cloud systems and patient records too. If one weak network gateway gets hacked, it could affect many connected devices at once. By looking at device attacks, network attacks and app attacks as separate boxes, the study risks missing how these attacks link together. This also undersells the need for system wide protections, like managing who has access, splitting networks into separate zones and having a solid plan ready for when something goes wrong.

1.Limits of AI based security claims

The survey talks about using machine learning and deep learning to spot unusual activity and figure out what kind of threat it is. Think of these as smart computer programs that learn to notice patterns, patterns that simple, fixed rules might miss. But just because a program scores well in tests doesn't mean it will actually work well in a real hospital. The results

depend on what data was used to train it, what kinds of attacks it was tested against, whether the test data was balanced and how the testing was done.

Here's the problem: if a program is only tested on a small or made up set of data, it might fail badly in a real hospital, where the devices and network traffic look completely different. If the system cries wolf too often (false alarms), it will wear out hospital staff who have to check every alert. But if it misses real threats (false negatives), staff might feel falsely safe, not realizing danger is present.

The survey praises these smart detection systems, but it also admits they can be tricked. If someone sneaks bad data into the training process, messes with sensor readings, or exploits the way these models slowly drift and change over time, then the detection system itself becomes a target, a new way in for attackers.

So we shouldn't just ask does this system catch threats well? We also need to ask: can it h and le attacks aimed at itself? Can people understand why it made a decision? Does it protect privacy? How are updates h and led? and what happens when it gets something wrong? Human supervision and a backup plan matter a lot here, especially when an automatic alert could affect a patient's care.

2.Privacy and technology tradeoffs

The survey talks about federated learning and other methods to keep patient data private. Here's what that means in plain terms: instead of sending all patient data to one central place, each device or hospital keeps its own data and only shares updates about what it learned. This can help protect privacy, but it's not a perfect fix.

Here's the problem: even those updates can accidentally reveal private information, almost like how you might guess what someone ate for dinner just from the smell on their clothes. Also, spreading data across many different locations instead of one place creates new headaches. You need extra systems to keep everything coordinated and secure.

There's a similar issue with anonymizing data, which means removing names and details so people can't be identified. The problem is that if you combine an anonymized dataset with other information, people can sometimes be identified again, like piecing together a puzzle from scattered clues. So when someone claims a system protects privacy, they need to show exactly what risks they tested for and prove it with real evidence, not just describe the technology they used.

The survey is also a bit too optimistic about blockchain. For those who don't know, blockchain is basically a shared digital record book that many computers keep copies of, making it hard for anyone to secretly change past entries. This does make it easier to track who changed what and spot suspicious activity. But here's the catch: blockchain only guarantees that once data is recorded, it can't be secretly altered. It does nothing to check if the data was correct in the first place. Garbage in, still garbage, just permanently locked in.

Blockchain also brings its own set of problems, like how to store all that data, who gets to control it, privacy concerns and the fact that it can run slowly. In healthcare specifically, this

permanence becomes a real problem. Patient records sometimes need to be corrected or deleted, think of fixing a wrong diagnosis or removing data when a patient requests it. But blockchain is designed to make records permanent and unchangeable, which directly conflicts with that need.

So blockchain should be seen as a specialized tool, useful for specific situations where you need to track and verify changes, not as some all purpose fix for securing medical devices and health data systems.

3.Evidence and methodological concerns

The study is a careful review of existing research. It looked at about 500 results and picked 163 of them to study closely. That's a good starting point, but it's hard to fully copy or check their methods. Here's why: the words they used to search were very broad, they searched a huge number of databases and they didn't clearly explain how they picked which studies to keep or throw out. To make this easier to check in the future, they should share the exact search terms used for each database, the exact dates they searched, how many studies they removed and why and a full check on how good each study was.

Some of the comparisons in the paper also need a closer look. The paper gives scores for things like how well something works, how much extra computer power it needs and how well it scales up for bigger use. But these scores can be based on opinion unless the paper clearly explains the rules used to score things and shows the proof behind each score. The same problem shows up with percentages used to compare how well different defenses work against attacks. These numbers often come from different tests, different setups and even different ideas of what working well means. So comparing them side by side, like in a chart, can look neat and exact, even though the numbers underneath aren't really measuring the same thing.

The paper's ideas for the future, like creating standard rules, saving energy, making AI easier to understand , keeping systems secure over their whole lifespan and following regulations, are all important topics. But they would be more useful if the authors ranked them by how urgent they are and explained exactly how to measure success for each one. For example, they could judge new security defenses by checking things like how fast they respond, how much energy they use, how well they catch real attacks, how quickly they recover after a problem, whether they leak private information and how they affect the daily work of doctors and nurses.

Positive and Negative Aspects of IoMT Security and Privacy

The Internet of Medical Things, or IoMT, is a simple idea. It means medical devices, communication networks and healthcare apps are all connected to each other, kind of like how your phone, smartwatch and laptop can all talk to one another.

This connection can really help patients. Doctors can watch a patient's health all the time, not just during checkups. and they can get important information faster when they need it.

Connecting all these devices also opens the door to new problems. Hackers could break in and steal private health information. Patient privacy could be at risk. and if something goes wrong with the system, it could disrupt how hospitals and clinics actually run.

Positive aspects

1.Continuous and remote care: Wearable gadgets and devices that check on you from a distance can track things like your heart rate and breathing. They let doctors keep an eye on patients even when they're not in the hospital. This means doctors can spot problems sooner and act faster.

2.More informed decisions: Getting the right data quickly can help doctors figure out what's wrong, plan how to treat it and give care that fits each person's needs.

3.Improved efficiency: When hospital systems are connected to each other, they can work together better. This means doctors and nurses can do their jobs more easily, hospitals can run more smoothly and some patients might not need to come back for extra visits they didn't really need.

4.Early detection and prevention: Looking at data closely can help spot troubling patterns early, before they turn into bigger problems.

5.Potential for privacy conscious analytics: New methods let different organizations study patient data together without having to pool all the raw data in one place. Each group keeps its own data private at home, but they can still learn from each other.

6.Stronger security tools are emerging: Lightweight cryptography, secure device authentication, AI assisted monitoring and secure update mechanisms can help protect devices and data when appropriately implemented.

Negative aspects and challenges

1.Exposure of sensitive health data: Patient information can travel through many places. It might pass through devices, networks, connection points and online storage systems. If the security on any of these is weak, like a lock that's easy to pick, it can lead to information leaks or people seeing data they shouldn't see.

2.Threats to patient safety: If a medical device gets tampered with, hacked, or stops working properly for a while, it can lead doctors to make the wrong call. That could end up hurting the patient.

3.Many points of attack: Someone could physically tamper with a device. Someone could intercept or disrupt data as it travels over a network. and apps could be attacked using ransomware (software that locks your files until you pay), phishing (fake messages that trick you into giving away private information), or software exploits (taking advantage of weaknesses in a program to break in).

4.Limited device resources: Some medical devices connected to the internet don't have much computing power, memory, or battery life. Think of it like trying to run a powerful security system on an old, weak phone. Adding strong security to these devices can slow them down or drain their battery faster.

5.AI systems are not automatically reliable: Machine learning tools can make mistakes. They might sound an alarm when nothing is wrong. They might miss a br and new type of attack they haven't seen before. Someone could even trick them by feeding them bad or misleading data on purpose. So just because a tool claims to be highly accurate doesn't mean it will actually work well in the real world.

6.Interoperability and maintenance difficulties: Devices from different manufacturers may use incompatible standards. Long device lifetimes also complicate patching, upgrades and vulnerability management.

7.Cost and complexity: Keeping things safe takes money, skilled people, constant watching and teamwork when something goes wrong. Because of this, staying protected all the time can be hard.

8.Tradeoffs in proposed technologies: The trade offs of new technology: Blockchain is a kind of digital record book that many people can check, so it can make it easier to track where information came from. But it has limits. It does not check if the information put into it is true in the first place. Think of it like a notebook that never gets erased. If someone writes something wrong in it, that mistake stays there forever, clear for everyone to see, but still wrong. Blockchain can also slow things down and make them harder to manage. Methods that protect people's privacy also need careful planning and lots of testing before they can be trusted.